1. Scrape Target exposes TimeCrypt encrypted metric values at /metrics endpoint

2. Scrape Target encrypts (label,value) under a generic metric name: enc_series & enc_series_timeid

(label, value) pairs: [
(__name__, enc_series_timeid),
(instance, localhost:2112),
(job, myapp)
(sid, fd862a389a21eb9c)
]

data: [1784745958.910, 12]
(data: [timestamp, TimeCrypt_time_index])

(label, value) pairs: [
(__name__, enc_series),
(instance, localhost:2112),
(job, myapp)
(sid, fd862a389a21eb9c)
]

data: [1784745958.910, 23432]
(data: [timestamp, TimeCrypt_encrypted_float_value])

/metrics

3. Generate TimeCrypt keystream for encrypting values

Go Application (myapp)

4. Replace Prometheus index with writer encrypted **(label,value)** pair as **keyword** and **sid** as document id (timeseries id)

Scrape Target

(label, value) pairs: [
(__name__, myapp_processed_ops_timeid),
(instance, localhost:2112),
(job, myapp)
]

data: [1784745958.910, 12]
(data: [timestamp, TimeCrypt_time_index])

(label, value) pairs: [
(__name__, myapp_processed_ops),
(instance, localhost:2112),
(job, myapp)
]

data: [1784745958.910, 23432]
(data: [timestamp, TimeCrypt_encrypted_float_value])

/metrics

2. Generate TimeCrypt keystream for encrypting values

Go Application (myapp)

Scrape Target

(label, value) pairs: [
(__name__, myapp_processed_ops),
(instance, localhost:2112),
(job, myapp)
]

Encrypt (label, value) pair as keyword to an epoch + writer

PRF Keyed with Writer Key

enc((__name__, myapp_processed_ops))

enc((instance, localhost:2112))

enc((job, myapp))

sid
(document/timeseries id)

HERMES Encrypted Index added to Prometheus

5. Prometheus scrapes the endpoint at regular intervals (say 10s)

HERMES Inverted Index

enc(labxx=valxx)	<list of sids>
enc(status="200")	1 2 5 99 1000 1001
enc(method="GET")	2 3 4 5 6 9 10 1502
Intersect	2 5 1502

Prometheus

6. Extract plaintext metrics names, (label,value) pairs from expression

7. Create search tokens out of (label,value) pairs using HICKAE aggregate keys

8. Query encrypted (label,value) pairs using the search tokens

enc_series {__hermes__="q0,q1"}[window]

9. Prometheus finds time series based on encrypted keywords and returns the TimeCrypt encrypted values

10. Derive keys from TimeCrypt's KDF tree to decrypt values

11. Run complete expression on decrypted values and fire alerts

delta(myapp_processed_ops[1m]) < 0
every 15s

Alert Rule Evalutor

12. Send notifications

Alert Manager